

WRRRA M 0 12 Proper Time Updates and Mode Spectra under Finite Boundaries

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Abstract

The five calibrated inputs and electron mode 23 of 0.11 are frozen while shutter updates are assigned to the proper time of each timelike worldline. Flat, accelerated, finite weak-gravity and uniform-expansion clocks are executed together with finite fold and spatial spectra. Integer branches required to recover energy from update phases are disclosed. Continuous free dispersion and boundary-discretized modes are both calculated. Fixed-volume updates of the inherited SI energy ledger connect through an exact time conversion; the old slow expansion schedule fails identification with that SI phase clock. The failure and the scope of completed connections remain explicit.

Verification inputs

The complete 0.11 input is frozen at SHA256

67de28abf0741c406178a2fe4ce07962e69bce827c3196b78e887da3bd0dab13. $G=6.67430 \times 10^{-11}$ SI, $H_0=67.4$ km/s/Mpc, energy shares 0.0493/0.265/0.6857 and electron rest energy 510998.95069 eV are retained.

Arithmetic address shares 0.05/0.268/0.682 belong to a separate ledger. $c=299792458$ m/s, $h=6.62607015 \times 10^{-34}$ J s and $1 \text{ eV}=1.602176634 \times 10^{-19}$ J are inherited; no new constant is fitted.

New constitutive inputs are phase interval $\epsilon=0.05$, 128 finite modes, fold boundary phases $\eta=0/0.5$, periodic/Dirichlet spatial boundaries and $L=20 \ell_\mu$. The accelerated $\beta(s)=0.2+0.5s$ path is prescribed, rather than derived from gravitational equations of motion. The $R=200$ kpc static patch and conditional $\Phi=\Psi$ are inherited. Pages 5, 11 and 12 of upper hypothesis r1 supply the worldline-update and phase-energy-branch provenance. Epsilon, boundaries and length are choices, not a measured minimum time or identified physical particle confinement.

WRRRA-specific transformation

The electron calibration supplies $\mu_E=E_e/23$, from which time/length units and the epsilon-scaled proper-time step follow. ℓ_μ is a unit reference. The microscopic cavity length is not assigned to cosmic size or the undetermined 0.11 L_0 . Planck time is a comparison scale; $c \Delta \tau$ does not prove an independent minimum spatial length.

$$\mu_E = E_e / 23, t_\mu = \hbar / \mu_E, \ell_\mu = c t_\mu, \Delta \tau_* = \epsilon t_\mu, \epsilon = 0.05. \quad (1)$$

Quantity	Calculated value	Unit
μ_E	22217.345682174	eV
t_μ	2.962603932832e-20	s
$\Delta \tau$	1.481301966416e-21	s
L_{cavity}	1.776332630208e-10	m

The execution gives delta_tau=1.481301966416e-21 s and delta_tau/t_P=2.747605735737e+22. Doubling epsilon leaves the energy spectrum and electron calibration unchanged.

A timelike path uses accumulated proper time. In the weak static patch $A=1+2\Phi/c^2$ and $B=1-2\Phi/c^2$, beta_local is speed measured by a static observer, rather than the coordinate speed. $A=B=1$ in flat space. Different paths are not forced to share one global shutter tick.

$$\begin{aligned} c^2 d\tau^2 &= A c^2 dt^2 - B dx^2, \\ \beta_{local}^2 &= \frac{B}{A c^2} \left(\frac{dx}{dt} \right)^2, \frac{d\tau}{dt} = \sqrt{A(1 - \beta_{local}^2)}. \end{aligned} \quad (2)$$

$$\tau(t) = \int_0^t \frac{d\tau}{dt'} dt', k = \lfloor \tau / \Delta\tau_* \rfloor, r_\tau = \tau / \Delta\tau_* - k, \tau(t_k) = k \Delta\tau_*. \quad (3)$$

For coordinate interval $T=64 \Delta\tau$, beta=0/0.6/0.8 gives 64/51/38 complete updates with a separate residual phase interval. The overlap column concerns an equally weighted fold-mode pair separated by μ_E ; it is not an outcome probability or physical record. A null path has zero proper time, but no rest-clock update count is assigned. Photon propagation is not declared frozen.

beta	tau / delta_tau	Complete ticks	Coherent overlap
0.0	64.0	64	0.000852612
0.6	51.2	51	0.082205611
0.8	38.4	38	0.328925174

$$t' = \gamma_u(t - ux/c^2), x' = \gamma_u(x - ut), c^2 \Delta t'^2 - \Delta x'^2 = c^2 \Delta t^2 - \Delta x^2. \quad (4)$$

The accelerated path is divided into 16/32/64/128 chords and each proper interval compared in a frame boosted by $u/c=0.3$. Errors against the path integral $\tau/T=0.877980386396$ decrease by approximately four per refinement. Solving accumulated proper time gives 57 events including the initial point $k=0$, representing 56 completed updates and a retained final residual interval. This is a numerical convergence check on a finite path.

The static local-clock potential integrates the inherited $g(r)$ to finite boundary R . $\Phi(R)=0$ normalizes the boundary clock. No reference at infinite distance is introduced. Independent numerical differentiation of the potential returns the same gravitational acceleration.

$$\Phi(r) = - \int_r^R g(s) ds, R = 200 \text{ kpc}, \Phi(R) = 0, \frac{v_R}{v_r} = \sqrt{A(r)/A(R)}. \quad (5)$$

r / kpc	Phi / c^2	Static d_tau / d_t
3	-1.728306260831e-06	0.999998271692
8.2	-1.248693279810e-06	0.999998751306
50	-4.966030243693e-07	0.999999503397
200	-0.000000000000e+00	1.000000000000

At 8.2 kpc, the inherited 207.510905127 km/s rotation gives a moving-clock $d_{\tau}/d_t=0.999998511748$. The uniform carrier I/N is stationary under any Hamiltonian, allowing FRW clock integration while retaining the uniform energy, pressure and expansion outputs. Constant peculiar $\beta=0.6$ specifies a path and assumes additional motion. Enormous update counts are floating-point order estimates, not exact integers or simulations of every event.

$$H(a) = H_0 \sqrt{(f_\phi + f_c) a^{-3} + f_b}, \Delta\tau = \int_{a_0}^{a_1} \frac{\sqrt{1 - \beta_{pec}^2}}{a H(a)} da. \quad (6)$$

A self-adjoint energy operator is now constructed on a finite fold basis. Every realized matrix has finite dimension $N=128$. F is the normalized Fourier matrix and η the declared fold boundary phase; allowed modes run from -64 through 63 . The construction uses a finite spectral projection of the derivative; grid error is not hidden by refitting the electron mode.

$$F_{jn} = N^{-1/2} e^{2\pi i j n / N}, H_{fold, \eta} = F \text{diag}(\mu_E |n + \eta|) F^\dagger, H = H^\dagger. \quad (7)$$

F represents the periodic-gauge field $\phi(y)$. The physical fold field $\psi(y) = \exp(2\pi i \eta y) \phi(y)$ encodes its boundary phase through the derivative shift $n + \eta$.

$$\psi(y+1) = e^{2\pi i \eta} \psi(y), \eta = 0 \text{ or } 1/2, E_{23,0} = 23 \mu_E = E_e. \quad (8)$$

$\eta=0$ reproduces the zero-intercept 0.11 mass branch. $\eta=0.5$ changes mode 23 to 522107.623531087 eV without recalibration. This is boundary sensitivity under the same calibration, not a claim that the physical electron boundary changed or a new particle identification. Equal ± 23 energies at $\eta=0$ do not determine charge sign; charge remains in the inherited filter ledger.

A separate finite spatial interval carries a free positive-energy operator with the electron rest energy. K is the positive squared-wave-number operator for the chosen boundary. A periodic boundary allows zero momentum; Dirichlet constrains both endpoint values to zero. These are explicit finite-domain boundary constraints, rather than a modeled physically infinite energy barrier.

$$K = F_B \text{diag}(k_n^2) F_B^\dagger, H_{space} = \sqrt{E_e^2 I + (\hbar c)^2 K}, L = 20 \ell_\mu. \quad (9)$$

$$k_n^{periodic} = 2\pi n / L, k_n^{Dirichlet} = \pi n / L, \psi_n(x) \propto \sin(\pi n x / L), n = 1, \dots, N. \quad (10)$$

Boundary / operator	Mode	Energy / eV	Kinetic / eV
fold periodic	0	0.000000000	—
fold periodic	23	510998.950690000	—
dirichlet	1	511010.867747384	11.917057384
dirichlet	2	511046.617252180	47.666562180
periodic	0	510998.950690000	0.000000000
periodic	1	511046.617252180	47.666562180

The table distinguishes total energy from kinetic energy after subtracting rest energy. It is not fitted to atomic lines or an observed cavity. Finite $L=20/40/80 \ell_\mu$ controls halve momentum spacing and approximately quarter the first kinetic-energy gap. These controls assume no physically realized infinite space.

Local free dispersion without a confinement boundary responds continuously to finite real momentum inputs. Noninteger $pc/\mu_E=0.1/0.5$ runs under the same shutter update. Discrete time alone does not discretize every momentum or energy. The independent finite-difference electron-mode control has errors $-5.2271\%/ -1.3225\%/ -0.3316\%$ at $N=128/256/512$ while keeping μ_E fixed.

$$E(p) = \sqrt{E_e^2 + c^2 p^2}, \text{ } preal, E_{23,N}^{FD} = \mu_E \frac{N}{\pi} |\sin(23\pi/N)|. \quad (11)$$

A proper-time update is the unitary of the same Hamiltonian. Eigenvalues of U are independently computed to read θ . At the default epsilon, fold and both spatial spectra agree with phase recovery to numerical precision. Phase alone leaves an integer energy branch ℓ undetermined, so the generator and branch must be disclosed.

$$U_* = e^{-iH\Delta\tau_*/\hbar}, U_* v_j = e^{-i\theta_j} v_j, 0 \leq \theta_j < 2\pi. \quad (12)$$

$$E_j = \frac{\hbar}{\Delta\tau_*} (\theta_j + 2\pi \ell_j), \ell_j \in \mathbb{Z}, E_{band} = 2\pi \hbar / \Delta\tau_*. \quad (13)$$

$$e^{-i(H + mE_{band}I)\Delta\tau_*/\hbar} = U_*, m \in \mathbb{Z}. \quad (14)$$

At $\epsilon=0.25$, 77 fold modes require nonzero integer branches. Discarding them produces a large energy-recovery failure. Reading U in the generator eigenbasis and adding the disclosed branches restores the energies. The branch integers are not newly inferred from U alone.

Outputs and comparison with the inherited clock

The 0.10 fixed-volume $E_{\text{star}}=ucrit$ V0 ledger and dimensionless ξ coordinate convert exactly to SI proper time using the same energy matrix. Microscopic fold/spatial spectra are separate probes; their electron energy is not added again to macroscopic density. $V0=1 \text{ m}^3$ is the declared collective ledger volume, rather than a volume deriving a fundamental particle clock.

$$U_{0.10}(\xi) = e^{-i\xi \hat{E}/E_*} = e^{-i\hat{E}\tau/\hbar}, \xi = E_* \tau / \hbar, \rho(\tau) = U \rho_0 U^\dagger. \quad (15)$$

States and D/R loads are executed at $k=0/1/2/4/8$ actual $\Delta\tau$ intervals. Total energy remains $3.200810499959 \times 10^{-10} \text{ J}$ while the D-load range is 0.038059728227. Normalization and positivity hold within numerical tolerances. This fixed-volume internal exchange includes no measurement apparatus or physical record generation yet. The final overlap expression in equation 16 applies only to an equal-weight fold-mode pair with gap μ_E ; it is not a general overlap formula for the collective ledger state.

$$Tr \rho = 1, Tr(\rho \hat{E}) = constant, |\langle \psi_0 | U(\tau) | \psi_0 \rangle|^2 = \cos^2(\mu_E \tau / 2\hbar). \quad (16)$$

However, the slow $\omega_{info}=0.7H0$ used in the 0.10 expansion-state schedule is not $E_{\text{star}}/\hbar$. The frequencies are $1.528999668760 \times 10^{-18} \text{ s}^{-1}$ and $7.272096256981 \times 10^{24} \text{ s}^{-1}$, respectively, with a ratio near 2.10×10^{-43} . Identifying the slow schedule as the physical SI phase of the same energy generator is falsified by this comparison. Retaining that schedule would require separate justification for another effective generator or interaction; this release does not claim to resolve it by substitution.

$$r_{clock} = \frac{\hbar \omega_{info}}{E_*} \neq 1, H_{eff} = r_{clock} \hat{E} \neq \hat{E}. \quad (17)$$

Computed comparison	Value	Unit / role
Default U eigenphase error	6.519258022308e-09	eV

Computed comparison	Value	Unit / role
Aliased naive error	8.664764816048e+05	eV
Declared-branch error	4.656612873077e-10	eV
Old / SI frequency	2.102556972196e-43	ratio

The old nonuniform slow expansion histories remain records of a constitutive schedule. The new physical phase is executed on the fixed-volume ledger; the uniform FRW clock connects through the stationary I/N state. Fully covariant nonuniform carrier/gravity/expansion dynamics are not closed here. Three positive finite steps check the error against the same Hamiltonian generator, and halving each step approximately halves that error. Equation 18 bounds the operator-norm error for finite Hermitian H without requiring physically infinite support.

$$H_{\delta} = \frac{i\hbar(U(\delta\tau) - I)}{\delta\tau}, \delta\tau > 0, \quad (18)$$

$$\|H_{\delta} - H\|_2 \leq \frac{\delta\tau}{2\hbar} \|H\|_2^2.$$

Falsifiers and reproduction

The connection must be revised if Lorentz transformations change proper time, the static-potential derivative fails to reproduce g, boundary energies and update phases disagree, integer branches are dropped or fixed-volume energy conservation fails. The failed identification of the old slow schedule with the SI phase is included in the verdict. All 36 new checks and 60 checks inherited from 0.11/0.10 pass. Invalid global/minimum-clock claims, unsupported infinite boundaries, out-of-patch paths and unexecuted physical-record inputs are rejected.

```
python -m pip install -r calculations/wrra_m_0_12/requirements.txt
python calculations/wrra_m_0_12/run_release.py
```

Parameters.json discloses inputs, provenance and constitutive choices. JSON and eight CSVs provide path clocks, update events, allowed modes, spatial dispersion and load exchange. A clean copy with captured results removed reproduces outputs and both manuscripts byte for byte. The 18 native Word equations and five computed tables are compared between editions, with every PDF page rendered and inspected. Numerical verification, document comparison and clean-copy reproduction verdicts are separate JSON records.

Reviewed 0.10 to 0.12 series DOI 10.5281/zenodo.23113101 · <https://doi.org/10.5281/zenodo.23113101>

Input SHA256 c80742fa4d0e0bdbb94e3390a2cc02da4a535168eaf66fb28f7318ec9b7b8b3d

Completion and subsequent scope

Under disclosed choices, 0.12 completes worldline proper-time updates, finite operator/boundary spectra and the fixed-volume SI-time conversion. The clock-identification failure and incomplete covariant connection are frozen alongside them. The origin or measured-minimum status of delta_tau and actual confinement boundary/length selection remain open. Version 0.13 executes physical binary outcomes, post-observation states and records; subsequent stages treat repeated uncertainty, stress, curvature, size, neutrinos and propagation. A new independent prediction or unique universe solution is not a prerequisite for this completion.

References

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Choi Wonsik. WRRRA_M Upper Two-Stage Filter Hypothesis 1.0-r1 (2026-10-02), pp. 5, 11–12; source contract in [references/clock_contract.md](#).

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